Appendix

1. Details of dataset.

The dataset used in this study spans both source and target domains, encompassing a total of 33 distinct tissue types, including Adrenal Gland, Bladder, Bone, Brain, Breast, Colon, Esophagus/Stomach, Eye, Fibroblast, Gallbladder, Hair, Head and Neck, Kidney, Leukemia, Liver, Lung, Lymphoid, Muscle, Myeloid, Nervous System, Other, Ovarian, Pancreas, Prostate, Sarcoma, Skin, Soft Tissue, Stomach, Testis, Thymoma, Thyroid, Uterine, and Vagina. Drug response profiles are available for nine drugs: 5-Fluorouracil, Cisplatin, Cyclophosphamide, Docetaxel, Doxorubicin, Etoposide, Gemcitabine, Paclitaxel, and Temozolomide.

This results in 297 possible tissue-drug combinations. However, only 232 combinations in the source domain and 91 in the target domain contain non-empty sample sets. Among these, 171 source-domain combinations and 30 target-domain combinations contain more than 10 samples. When the threshold is raised to 30 samples, only 74 source-domain and 22 target-domain combinations remain.

These statistics highlight the inherent imbalance in both tissue and drug coverage, with many combinations sparsely populated. Furthermore, each tissue-drug pair contains both sensitive and resistant samples, introducing additional granularity. This fine-grained heterogeneity, coupled with the challenge of transferring drug response knowledge across tissues from cell lines and patients, poses significant obstacles to precise cross-domain matching and robust model generalization.

The data can be downloaded from here and originates from publicly available sources not uploaded by the authors.

Datasets — https://drive.google.com/file/d/1azKGGPOCJNDP4vQbV1v8p_XqCrd7ekP1/view?usp=sharing

2. System configurations.

All experiments were conducted on a Linux-based workstation equipped with the following hardware and software specifications:

- CPU: AMD Ryzen 9 5950X
- Memory: 128 GB RAM
- GPU: NVIDIA RTX 4090D(24 GB memory)
- Framework: PyTorch version 2.4.1

3. Baseline configuration.

We summarize the hyperparameter settings used for all methods evaluated in this study, including our proposed MACB-DRP framework and baseline models. Where applicable, we follow the original implementations or reported configurations, with minor adjustments based on empirical performance or compatibility with our dataset.

- **MACB-DRP**: For MACB-DRP, we set the number of pre-training epochs to 300, fusion training epochs to 100, and contrastive learning epochs to 200. In the contrastive anchoring module, the number of top-K neighbors was fixed at $K = 7$, and the contrastive loss weight was set to $\alpha = 0.1$. All hyperparameters were kept consistent across all drugs.
- **TransDRP**: The original paper reports 200 epochs for pre-training and 300 epochs for adaptation. However, we observed improved performance using 100 pre-training epochs and 500 adaptation epochs, and thus adopted this configuration for the experiments.
- **WISER**: WISER appears to use different hyperparameter combinations for each drug. Based on the descriptions in the paper and appendix, we selected a configuration with strong overall performance and applied it uniformly across all drugs. Specifically, the model was pre-trained for 50 epochs and fine-tuned for 2500 epochs. The dropout rate was set to 0.0, disabling stochastic regularization, and the inverse temperature was set to 10 to sharpen the contrastive similarity distribution.
- **DruID**: We strictly followed the default parameters provided in the official codebase, including epochs = 50. However, DruID incorporates clinical-grade next-generation sequencing (cNGS) panels, which are not directly compatible with our dataset. To accommodate dimensional mismatches, we added a lightweight projection layer to the model. These adjustments, along with the domain-specific nature of the original data, led to suboptimal performance on our nine-drug benchmark.
- **CODE-AE**: The original paper does not specify the number of training epochs or related hyperparameters. Due to the similarity in code structure, we infer that WISER was developed based on CODE-AE. Accordingly, we adopted the same configuration as used for WISER: the model was pre-trained for 50 epochs and fine-tuned for 2500 epochs, with a dropout rate of 0.0.
- **VAEN**: We used the optimal parameters reported in the original paper. The model was trained for 100 epochs, and the dropout rate was set to 0.0, consistent with the settings used for WISER and CODE-AE.

4. Complete performance tables.

Results for individual drugs using MACB-DRP, TransDRP, WISER, DruID, CODE-AE, and VAEN are presented in Tables 1 to 6.

5. Visualization analysis.

Hierarchical visualization results for four drugs are shown in Figure 1.

Figure 2 presents the t-SNE visualization of the raw gene expression profiles from CCLE and TCGA before model training. The embeddings show a clear separation between the source and target domains. In terms of tissue types, CCLE samples appear intermixed, whereas TCGA samples exhibit more distinct tissue-specific clustering.

Figure 3 presents the t-SNE visualization of the feature embeddings learned by TranDRP after domain adaptation. The visualization shows that TranDRP achieves domain-level alignment but does not produce clear separation across tissue types or drug-related feature patterns.

Drugs	AUC	AUPR	F1	ACC
5-Fluorouracil	0.5936/0.0630	0.7106/0.0442	0.6009/0.0590	0.7007/0.0574
Cisplatin	0.6514/0.0517	0.8124/0.0418	0.6835/0.0656	0.7814/0.0645
Cyclophosphamide	0.7063/0.0302	0.9419/0.0146	0.7282/0.1151	0.8285/0.0875
Docetaxel	0.5220/0.0503	0.7148/0.0452	0.5484/0.0510	0.6422/0.0723
Doxorubicin	0.5791/0.0645	0.8019/0.0495	0.5673/0.0818	0.6579/0.0951
Etoposide	0.7864/0.0536	0.9329/0.0204	0.8333/0.0383	0.8983/0.0260
Gemcitabine	0.5883/0.0195	0.5916/0.0244	0.5609/0.0234	0.5895/0.0297
Paclitaxel	0.5744/0.0328	0.7491/0.0235	0.5919/0.0185	0.6919/0.0221
Temozolomide	0.6838/0.0387	0.5136/0.0614	0.7522/0.0097	0.1372/0.1575

Table 1: Drug response prediction results of MACB-DRP on 9 drugs.

Drugs	AUC	AUPR	F1	ACC
5-Fluorouracil	0.7091/0.1124	0.8050/0.0743	0.6100/0.0838	0.6249/0.1035
Cisplatin	0.6372/0.0562	0.8005/0.0400	0.6187/0.0585	0.7063/0.0669
Cyclophosphamide	0.6217/0.0791	0.9425/0.0134	0.6353/0.2324	0.7308/0.2268
Docetaxel	0.5854/0.1103	0.7478/0.0682	0.5937/0.1010	0.6575/0.1195
Doxorubicin	0.5792/0.1229	0.7932/0.0845	0.5522/0.1354	0.6327/0.1568
Etoposide	0.6298/0.0435	0.8657/0.0085	0.5813/0.0894	0.6858/0.0932
Gemcitabine	0.5588/0.0357	0.5415/0.0302	0.5522/0.0202	0.4320/0.0715
Paclitaxel	0.5813/0.0652	0.7782/0.0472	0.5288/0.0649	0.5861/0.0822
Temozolomide	0.6666/0.0526	0.4852/0.1099	0.5836/0.2440	0.3857/0.1086

Table 2: Drug response prediction results of TransDRP on 9 drugs.

Drugs	AUC	AUPR	F1	ACC
5-Fluorouracil	0.6194/0.0103	0.7002/0.0119	0.4957/0.2778	0.5336/0.0929
Cisplatin	0.6208/0.0116	0.7837/0.0117	0.8406/0.0001	0.7261/0.0038
Cyclophosphamide	0.5461/0.1012	0.9181/0.0288	0.9538/0.0000	0.9118/0.0000
Docetaxel	0.5300/0.0032	0.6664/0.0028	0.6791/0.0237	0.5862/0.0216
Doxorubicin	0.5082/0.1249	0.7411/0.0626	0.5070/0.4018	0.5384/0.2423
Etoposide	0.6773/0.0725	0.9169/0.0197	0.8827/0.0528	0.8021/0.0625
Gemcitabine	0.5814/0.0022	0.568/0.0124	0.6295/0.0441	0.5200/0.0386
Paclitaxel	0.5458/0.0106	0.7142/0.0205	0.7896/0.0065	0.6658/0.0115
Temozolomide	0.6055/0.0758	0.3869/0.0924	0.4210/0.0324	0.3493/0.1910

Table 3: Drug response prediction results of WISER on 9 drugs.

Drugs	AUC	AUPR	F1	ACC
5-Fluorouracil	0.4295/nan	0.5801/nan	0.3909/nan	0.2556/nan
Cisplatin	0.4203/nan	0.6657/nan	0.2854/nan	0.1337/nan
Cyclophosphamide	0.5708/nan	0.9326/nan	0.3647/nan	0.4757/nan
Docetaxel	0.5824/nan	0.7237/nan	0.4654/nan	0.4295/nan
Doxorubicin	0.4700/nan	0.7363/nan	0.3836/nan	0.4096/nan
Etoposide	0.3766/nan	0.7502/nan	0.1667/nan	0.1111/nan
Gemcitabine	0.4406/nan	0.4341/nan	0.4652/nan	0.3692/nan
Paclitaxel	0.6072/nan	0.7545/nan	0.3874/nan	0.2527/nan
Temozolomide	0.5338/nan	0.2958/nan	0.7239/nan	0.0513/nan

Table 4: Drug response prediction results of DruID on 9 drugs.

Drugs	AUC	AUPR	F1	ACC
5-Fluorouracil	0.5484/0.0238	0.6784/0.0170	0.3186/0.1808	0.4645/0.0517
Cisplatin	0.5755/0.0250	0.7945/0.0174	0.7972/0.0278	0.6782/0.0260
Cyclophosphamide	0.3553/0.0459	0.8874/0.0174	0.8870/0.0352	0.7988/0.0570
Docetaxel	0.4013/0.0219	0.6288/0.0113	0.5696/0.124	0.4742/0.1062
Doxorubicin	0.4668/0.0783	0.7713/0.0495	0.2277/0.1463	0.3358/0.0460
Etoposide	0.6542/0.0585	0.8897/0.0217	0.8577/0.0117	0.7563/0.0169
Gemcitabine	0.5561/0.0095	0.5632/0.0117	0.6300/0.0312	0.5052/0.0219
Paclitaxel	0.4758/0.0517	0.7061/0.0403	0.7996/0.0183	0.6685/0.0224
Temozolomide	0.3876/0.0610	0.2405/0.0355	0.4048/0.0000	0.2537/0.0000

Table 5: Drug response prediction results of CODE-AE on 9 drugs.

Drugs	AUC	AUPR	F1	ACC
5-Fluorouracil	0.5346/0.0208	0.6572/0.0129	0.5185/0.1388	0.4982/0.0543
Cisplatin	0.5021/0.0200	0.7274/0.0143	0.7980/0.0165	0.6763/0.0187
Cyclophosphamide	0.4773/0.0382	0.9001/0.0110	0.6949/0.3505	0.6388/0.2849
Docetaxel	0.5659/0.0898	0.7157/0.0418	0.6312/0.1768	0.5660/0.1107
Doxorubicin	0.3557/0.0353	0.6737/0.0145	0.3976/0.2518	0.3899/0.1829
Etoposide	0.4648/0.0689	0.8140/0.0239	0.7453/0.1112	0.6312/0.1180
Gemcitabine	0.5392/0.0145	0.5212/0.0102	0.5458/0.0265	0.5243/0.0207
Paclitaxel	0.4625/0.0226	0.6699/0.0093	0.7519/0.0152	0.6306/0.0142
Temozolomide	0.6421/0.0128	0.4356/0.0274	0.4126/0.0195	0.6060/0.0571

Table 6: Drug response prediction results of VAEN on 9 drugs.

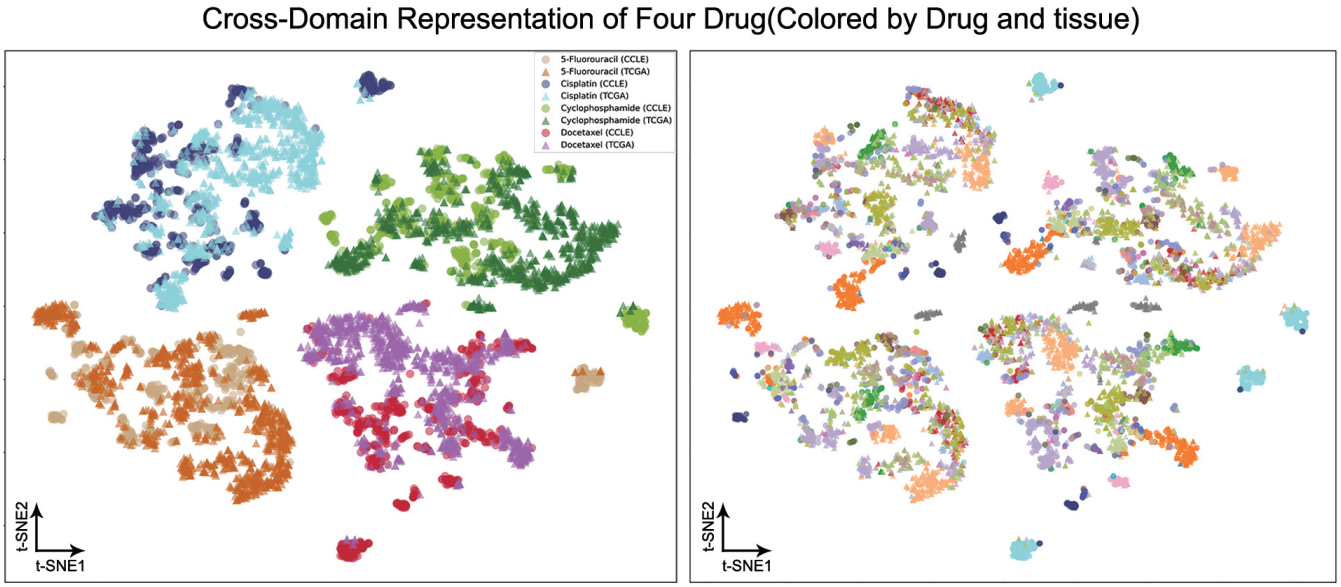


Figure 1: T-SNE visualizations of latent representations for four drugs, annotated by tissue type

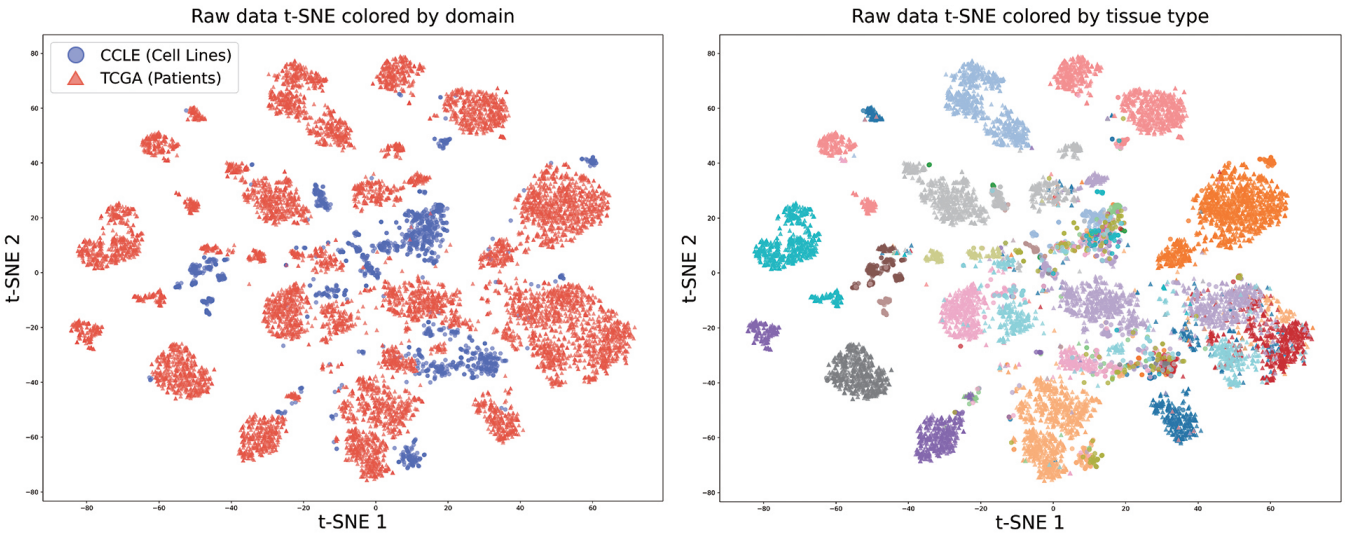


Figure 2: T-SNE visualization of raw gene expression data from CCLE and TCGA prior to training. The left panel is colored by domain (source: CCLE; target: TCGA), and the right panel is colored by tissue type.

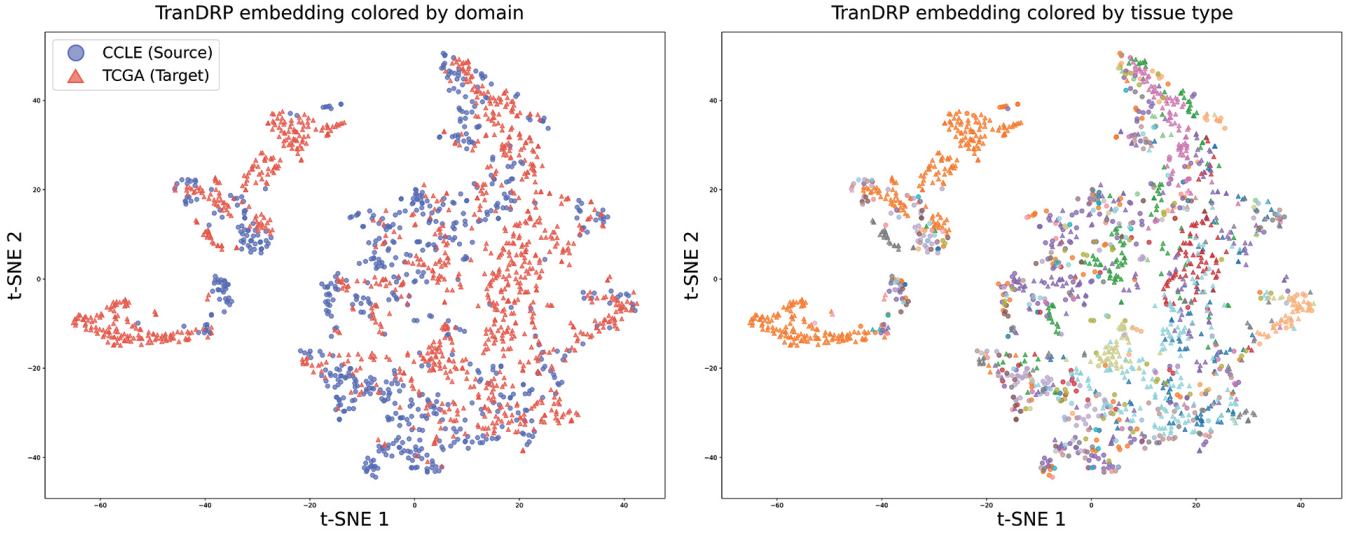


Figure 3: T-SNE visualization of TranDRP feature embeddings after domain adaptation. The left panel is colored by domain (source: CCLE; target: TCGA), and the right panel is colored by tissue type.